



**DEPARTMENT OF COMPUTER
SCIENCE AND ENGINEERING**

Rajiv Gandhi University of Knowledge Technologies
Nuzvid, Eluru, Andhra Pradesh – 521202

**Application for Predicting Blood
Cell Counts Using Deep Learning**

A Major Project Report
Submitted in partial fulfillment for the degree of

BACHELOR OF TECHNOLOGY
in
COMPUTER SCIENCE AND ENGINEERING
Submitted by

K NARASIMHA REDDY	(N190685)
K SHANMUKHA SAI VIGNESH	(N191122)
D ASWINI	(N190662)
Y SUPRIYA	(N190729)
CH TARUN	(N190851)

Under the Esteem Guidance of

Mr.KRISHNA KUMAR SINGH(Asst Prof)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Rajiv Gandhi University of Knowledge Technologies –
Nuzvid

Nuzvid, Eluru, Andhra Pradesh – 521202

CERTIFICATE OF COMPLETION

This is to certify that the work entitled, “**Android Application for Predicting Blood Cell Counts Using Deep Learning**” is the bonafide work of **K NARASIMHA REDDY** (ID No: N190685), **K SHANMUKHA SAI VIGNESH** (ID No: N191122), **D ASWINI** (ID No: N190662), **Y SUPRIYA** (ID No: N190729), **CH TARUN** (ID No: N190851) carried out under our guidance and supervision for the 3rd year project of Bachelor of Technology in the department of Computer Science and Engineering under RGUKT IIIT Nuzvid. This work was done during the academic session December 2025 – May 2025, under our guidance.

Mr.K Krishna Singh
Assistant Professor,
Department of CSE,
RGUKT Nuzvid.

Ms. S. Bhavani
Head of the Department,
Department of CSE,
RGUKT Nuzvid.

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Rajiv Gandhi University of Knowledge Technologies
Nuzvid, Eluru, Andhra Pradesh – 521202

CERTIFICATE OF EXAMINATION

This is to certify that the work entitled, “**Android Application for Predicting Blood Cell Counts Using Deep Learning**” is the bonafide work of **K NARASIMHA REDDY** (ID No: N190685), **K SHANMUKHA SAI VIGNESH** (ID No: N191122), **D ASWINI** (ID No: N190662), **Y SUPRIYA** (ID No: N190729), **CH TARUN** (ID No: N190851) and hereby accord our approval of it as a study carried out and presented in a manner required for its acceptance in final year of Bachelor of Technology for which it has been submitted. This approval does not necessarily endorse or accept every statement made, opinion expressed or conclusion drawn, as a recorded in this thesis. It only signifies the acceptance of this thesis for the purpose for which it has been submitted.

Mr.Krishna Kumar Singh
Assistant Professor,
Department of CSE,
RGUKT-NUZVID

Project Examiner
RGUKT-Nuzvid

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Rajiv Gandhi University of Knowledge Technologies
Nuzvid, Eluru, Andhra Pradesh – 521202

DECLARATION

K NARASIMHA REDDY (ID No: N190685), **K SHANMUKHA SAI VIGNESH** (ID No: N191122), **D ASWINI** (ID No: N190662), **Y SUPRIYA** (ID No: N190729), **CH TARUN** (ID No: N190851) hereby declare that the project report entitled “**Android Application for Predicting Blood Cell Counts Using Deep Learning**” done by us under the guidance of **Mr.Krishna Kumar Singh**, Assistant Professor is submitted for the partial fulfillment for the award of degree of Bachelor of Technology in Computer Science and Engineering during the academic session January 2024 - June 2024 at RGUKT-Nuzvid.

We also declare that this project is a result of our own effort and has not been copied or imitated from any source. Citations from any websites are mentioned in the references..

Date: 22-06-2024	K NARASIMHA REDDY	(N190685)
Place: Nuzvid	K SHANMUKHA SAI VIGNESH	(N191122)
	D ASWINI	(N190662)
	Y SUPRIYA	(N190729)
	CH TARUN	(N190851)

ACKNOWLEDGEMENT

We would like to express our heartfelt profound gratitude and deep regards to our guide **Mr.Krishna Kumar Singh** for his exemplary guidance, monitoring, and constant encouragement to us throughout the B.Tech course. We shall always cherish the time spent with him during the course of this work due to the invaluable knowledge gained in the field of reliability engineering.

We are extremely grateful for the confidence bestowed in us and entrusting our project entitled “**Android Application for Predicting Blood Cell Counts Using Deep Learning**”.

We express gratitude to **Dr. S. Bhavani** (HOD of CSE) and other faculty members for being a source of inspiration and constant encouragement which helped us in completing the project successfully.

Our sincere thanks to all the batch mates of **2019 CSE**, who have made our stay at **RGUKT-NUZVID**, a memorable one.

Finally, yet importantly, we would like to express our heartfelt thanks to our beloved God and parents for their blessings, our friends for their help and wishes for the successful completion of this project.

ABSTRACT

This project aims to leverage machine learning techniques for the automated counting and identification of RBCs, WBCs, and platelets from microscopic blood images using the YOLO model. The primary objective is to develop a application that can predict blood cell counts quickly and affordably, even in regions lacking advanced laboratory facilities. This application seeks to democratize healthcare by enabling early disease detection and promoting healthier outcomes globally.

Utilizing the BCCD Dataset, the YOLO model is employed to count RBCs, WBCs, and platelets, providing a rapid and precise tool for hematological analysis. This method demonstrates significant improvements in efficiency and accuracy over traditional manual counting methods[1]. Results indicate that the YOLO model effectively identifies and quantifies blood cells, underscoring its reliability for medical diagnostics. Overall, this app enhances the efficiency of blood cell counting, presenting substantial potential benefits for healthcare applications, particularly in improving access to diagnostic tools and facilitating timely medical interventions.

Keypoints: Feature Extraction,Data Preprocessing, Machine Learning Model: YOLO, Libraries used in the project., TensorFlow, NumPy, Matplotlib, Pandas.

Contents

1	Introduction	5
2	Related Works	6
3	Counting of Blood Cells Using Manual Method	7
3.1	Purpose	7
3.2	Key Innovation	7
3.3	Limitation	7
4	Proposed Approach: Application for Predicting RBC, WBC, and Platelets Count with Microscopic Blood Image Using YOLO Model	8
4.1	Purpose	8
4.2	Innovative Approach	8
4.3	Significance	8
4.4	Performance Metrics	8
4.5	Smoother Convergence	8
5	Procedure	9
5.1	Step-1: Initialization	9
5.2	Step-2: Data Preprocessing	9
5.3	Step-3: Model Definition	9
5.4	Step-4: Training Loop	9
5.5	Step-5: Output Model	9
5.6	Model Architecture	10
6	Experiments - Datasets Used	10
6.1	Libraries Used	10
7	Formulas Used :	11
7.1	CIoU(Complete Intersection over Union) Loss:	11
7.2	DFL(Distribution Focal Loss)	11
8	System Hardware	11
9	Sample Code	12
9.1	Prediction	12
9.2	Train and Validation Loss Graph	13
10	Streamlit Blood cellcount Application Overview Code Implementation	13
10.1	Step 1: Importing Required Libraries	13
10.2	Step 2: Load YOLO Model	13
10.3	Step 3: Set App Title and Description	14
10.4	Step 4: Upload Image	14
10.5	Step 5: Perform Inference	14
10.6	Step 6: Display Detection Image	14
10.7	Step 7: Display Summary Results	14

11 Results and Discussions	15
11.1 Input Microscopic Blood Image	15
11.2 Detection and Highlighting Blood Cells	15
11.3 Summary of Detection	16
11.4 Performance Metrics	16
11.5 Confusion Matrix	16
12 Conclusion	17
12.1 Summary of App's Contributions	17
12.2 Advancements over Traditional Methods	17
12.3 Future Potential in Medical Diagnostics and AI Applications	17
13 Future Enhancements	19
13.1 Areas for Future Research and Development	19
13.2 Potential Applications and Extensions	19

1 Introduction

In the realm of healthcare, timely and accurate blood cell counts are critical for diagnosis and treatment. Traditional methods of blood cell analysis often require advanced laboratory facilities and manual intervention, which can be both costly and time-consuming. To address these challenges, we propose an application that leverages the YOLO (You Only Look Once) model for real-time analysis of blood samples[2]. This app is designed for emergency situations where quick blood cell counts are essential for making rapid treatment decisions. It can also be utilized by scientists to study diseases and health trends and by patients to monitor their health regularly. By automating the identification and quantification of RBCs (Red Blood Cells), WBCs (White Blood Cells), and platelets, the app aims to reduce the cost and minimize wastage associated with traditional Complete Blood Count (CBC) tests. The primary objective of this project is to enhance the accuracy and efficiency of blood cell counting through deep learning techniques. By training the YOLO model with large datasets, we aim to provide a reliable and automated system for blood cell analysis, which will improve diagnostic capabilities and overall healthcare outcomes.

Understanding the role of blood cells is crucial for this application:

Red Blood Cells (RBCs) transport oxygen from the lungs to the body's tissues and carry carbon dioxide back for exhalation. They are flat, disk-like microscopic cells[2].

White Blood Cells (WBCs) are part of the immune system and respond to injury or illness. These colorless cells have a round shape with a distinct nucleus[5].

Platelets play a vital role in preventing and stopping bleeding. They are tiny, anucleated discoid cells with an average diameter of 2–3 μm and a lifespan of about 10 days.

By integrating deep learning into the analysis of microscopic blood images, this app aims to provide a fast, accurate, and cost-effective solution for hematological diagnostics.

2 Related Works

Title	Author	Published Year	Key Concepts
Detection, classification, and counting blood cells using YOLOv8 [6].	Duong Trong Luong; Do Thanh Tuan; Dao Duy Anh; Tran Thuy Hanh; Hoang Thi Lan Huong; Tran Xuan Thang	2023	In this research, we propose to use YOLOv8 to detect, classify, and count normal white blood cells.
Examination of blood samples using deep learning and mobile microscopy [7].	Juliane Pfeil, Alina Nechyporenko, Marcus Frohme	2022	Training and optimisation of different deep learning methods for instance segmentation are used to detect and count the different blood cells.
Platelet Detection Based on Improved YOLO_v3 [8].	Platelet Detection Based on Improved YOLO_v3RENTING LIU, CHUNHUI REN, MIAOMIAO FU , ZHENGKANG CHU, AND JIUCHUAN GUO	2022	Experiments for platelet detection based on commonly used object detection model, such as You Only Look Once_v3 (YOLO_v3)
Habitual detection and measurement of human blood cells on hyperspectral imagery for convolutional neural network [9].	T. Arumuga Maria Devi and PaulRaj Thangaselvi	2024	YOLO structure has been set up with a changed arrangement BCCD Dataset of blood smear picture to normally recognise and check red platelets, white platelets, and platelets.
Identification and counting white blood cell subtypes and with CNN [10].	Singgih Bektı Worsito	2021	Recognizing and counting WBC subtypes with neural networks constructed from augmented data enrichment.

Figure 1: Realted works

3 Counting of Blood Cells Using Manual Method

3.1 Purpose

The manual method of counting blood cells is used to determine the number of different types of blood cells in a given volume of blood. This technique is crucial for diagnosing and monitoring various medical conditions, such as infections, anemia, and hematologic diseases. It provides detailed information about the patient's blood cell counts, which is essential for medical decision-making.

3.2 Key Innovation

The key innovation in the manual method is the use of a hemocytometer, a specialized counting chamber that allows for the precise enumeration of blood cells. The hemocytometer, combined with a microscope, enables laboratory technicians to manually count the cells in a defined volume of diluted blood. This approach ensures direct visualization and accurate counting of each cell type, including red blood cells (RBCs), white blood cells (WBCs), and platelets.

3.3 Limitation

Despite its utility, the manual method of counting blood cells has several limitations. It is time-consuming and labor-intensive, requiring skilled technicians to perform the counts accurately. The method is also prone to human error, leading to potential variability in results. Furthermore, the manual counting process can be tedious, especially for large sample volumes, and is not feasible for high-throughput testing. These limitations highlight the need for automated counting methods in clinical laboratories to ensure consistent and reliable results.

4 Proposed Approach: Application for Predicting RBC, WBC, and Platelets Count with Microscopic Blood Image Using YOLO Model

4.1 Purpose

The application for predicting RBC, WBC, and platelets count using microscopic blood images with the YOLO model is designed to automate the process of blood cell counting and classification[5]. This tool aims to assist medical professionals by providing a quick, accurate, and efficient method to analyze blood samples, thereby reducing the time and effort required for manual counting and minimizing human error.

4.2 Innovative Approach

The innovative approach of this app involves the utilization of the YOLO (You Only Look Once) model, a state-of-the-art object detection algorithm. The YOLO model is applied to microscopic images of blood smears to detect and classify different types of blood cells. By leveraging the speed and accuracy of YOLO, the app can process images in real-time, providing immediate results for RBC, WBC, and platelet counts[4].

4.3 Significance

The significance of this application lies in its potential to revolutionize hematology diagnostics by automating the labor-intensive process of blood cell counting. By providing accurate and rapid analysis, the app can significantly enhance the efficiency of clinical workflows, reduce the burden on laboratory technicians, and improve patient outcomes. Additionally, the application of the YOLO model in this domain showcases the versatility and power of deep learning techniques in medical imaging and diagnostics, paving the way for further advancements and innovations in healthcare technology.

4.4 Performance Metrics

Precision and Recall: Key performance metrics for the app include precision and recall rates for detecting RBCs, WBCs, and platelets. High precision ensures that the detected cells are correctly classified, while high recall indicates that most of the actual cells present in the image are detected. **Processing Speed:** Another critical metric is the processing speed of the app, measured in frames per second (FPS). This determines how quickly the app can analyze blood smear images and provide results. **Accuracy:** The overall accuracy of the app in predicting the counts of RBCs, WBCs, and platelets is evaluated against manual counting methods to ensure reliability.

4.5 Smoother Convergence

The use of the YOLO model ensures smoother convergence during the training process due to its unique architecture that performs both detection and classification in a single

forward pass. This integrated approach minimizes the computational complexity and enhances the model's ability to converge quickly and reliably, leading to more consistent and accurate predictions.

5 Procedure

5.1 Step-1: Initialization

The first step involves setting up the environment and importing the necessary libraries to facilitate the implementation of the to detect the blood cell types in microscopic images . This includes loading and preprocessing the bccd dataset, which is divided into training, validation, and test sets.

5.2 Step-2: Data Preprocessing

In this step, the input images and text files from the bccd dataset are used to create a data.yaml file to input for the models. Before that we divided the dataset into train ,test and valid dataset.

5.3 Step-3: Model Definition

This step involves how the YOLO model works. The YOLO model replaces the predefined 80 classes with our 3 classes those classes are WBC, RBC, Platelets.

5.4 Step-4: Training Loop

The original implementation of the YOLO configuration was trained for 80 different classes. To adopt it for blood cells identification, we modify it for three classes consisting of WBC, RBC, and platelets. Due to modifying the class number, the number of filters in the final convolutional layer in the CNN architecture is needed to be changed as well. YOLO predicts five values along with class probabilities for each anchor box. The values are the probability of having an object in a grid cell, x and y coordinates of the object, height, and width of the object. The model is trained with 30 epochs and batch size is 48 with initial learning rate 0.01.

5.5 Step-5: Output Model

After training the model with our datasets and our three classes ,The Yolo creates a model and store it in the runs folder with the name best.pt .We use that best.pt model for the application development.When we test the model with test images it will generate different loss graphs that are stored in runs folder.

5.6 Model Architecture

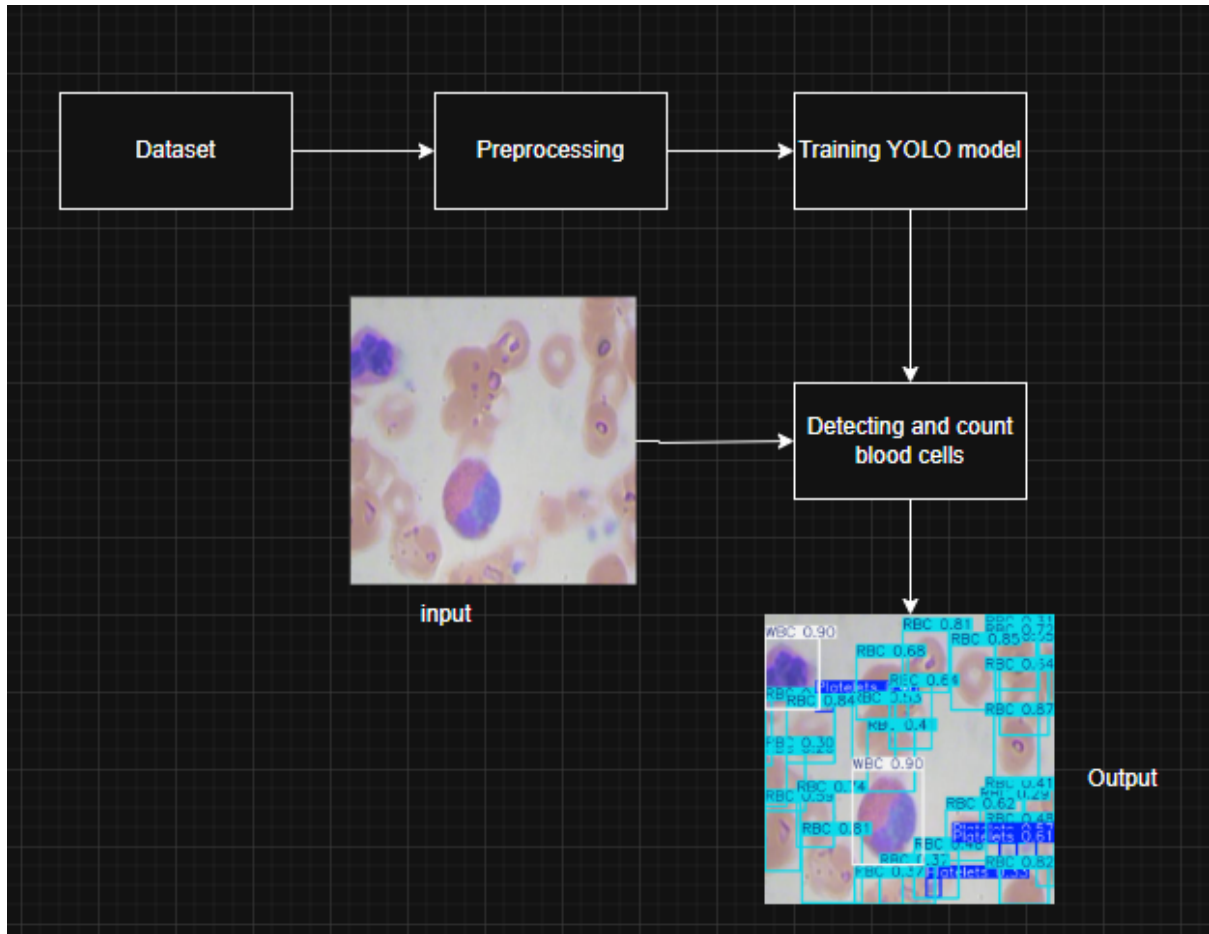


Figure 2: Model Architecture

6 Experiments - Datasets Used

6.1 Libraries Used

- **Ultralytics:** Ultralytics is the organization behind the popular YOLO (You Only Look Once) series of real-time object detection models. Their software packages and libraries are designed to facilitate easy use of these models, enabling developers to integrate object detection into various applications.
- **NumPy:** Libraries for numerical computations, and mathematical functions, essential for data preprocessing and augmentation.
- **Cv2 and Matplotlib:** Libraries for image processing and visualization, used for handling input images and visualizing generated samples and training progress.
- **Flet:** Flet is an open-source Python library that allows developers to build interactive web, desktop, and mobile applications without needing to know HTML, CSS,

or JavaScript. It simplifies the process of creating UI applications by using Python to define the UI components and their behavior.

7 Formulas Used :

7.1 CIoU(Complete Intersection over Union) Loss:

$$\text{CIoU}(b, b^{gt}) = 1 - \text{IoU}(b, b^{gt}) + \frac{\rho^2(b, b^{gt})}{c^2} + \alpha v \quad (1)$$

where:

- $\text{IoU}(b, b^{gt})$ is the Intersection over Union between the predicted box b and the ground truth box b^{gt} .
- $\rho(b, b^{gt})$ is the Euclidean distance between the central points of b and b^{gt} .
- c is the diagonal length of the smallest enclosing box covering both b and b^{gt} .
- v is a measure of the similarity of the aspect ratios of b and b^{gt} .
- α is a positive trade-off parameter, defined as $\alpha = \frac{v}{(1 - \text{IoU}(b, b^{gt})) + v}$.

7.2 DFL(Distribution Focal Loss)

$$\text{DFL}(p, q) = - \sum_{i=1}^K q_i \log(p_i) \quad (2)$$

where:

- p is the predicted probability distribution over K bins (e.g., object classes).
- q is the ground truth probability distribution over K bins, often a one-hot encoded vector.

8 System Hardware

Google Colab TP4 GPU

RAM Used: 7.66 GB/12.67 GB

Disk Used: 29.10 GB/78.19 GB

9 Sample Code

```

v Ultralytics

For using yolo predefined models we need to install the Ultralytics package using pip cmd

[ ] !pip install ultralytics

v Packages importing

from ultralytics import YOLO
import matplotlib.pyplot as plt
import numpy as np
import cv2

v Training

Using the command line interface
we need to pass the data in yaml format and epochs and image size and required parameters

[ ] !yolo task=detect mode=train model=yolov8s.pt data=/content/drive/MyDrive/MFDS/n/data.yaml epochs=30 imgsz=416

Training using python scripts we need to run just any one of the train step either cmd line or script

[ ] # Load a model
model1 = YOLO('yolov8s.pt')
results = model1.train(data='/content/drive/MyDrive/MFDS/n/data.yaml', epochs=10, imgsz=416) # Train the model

model=YOLO("/content/drive/MyDrive/MFDS/n/runs/detect/train/weights/best.pt")

results1 = model('/content/drive/MyDrive/MFDS/n/ok.jpg') # return a list of Results objects

# Process results list
for result in results1:
    boxes = result.boxes # Boxes object for bounding box outputs
    result.save(filename='result.jpg') # save to disk

```

9.1 Prediction

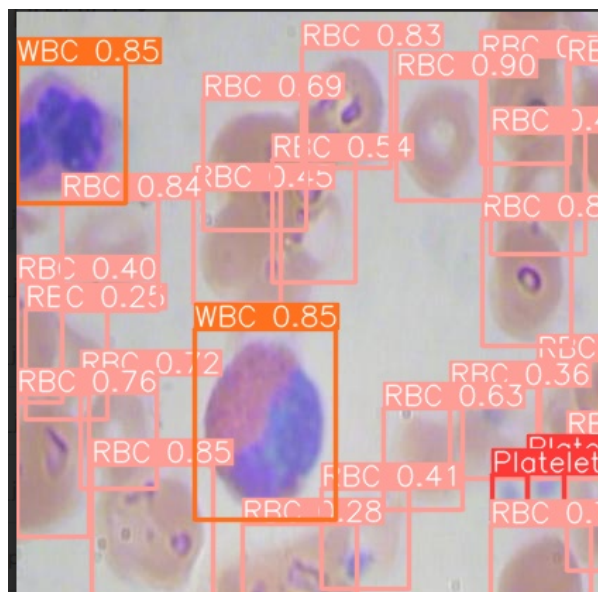
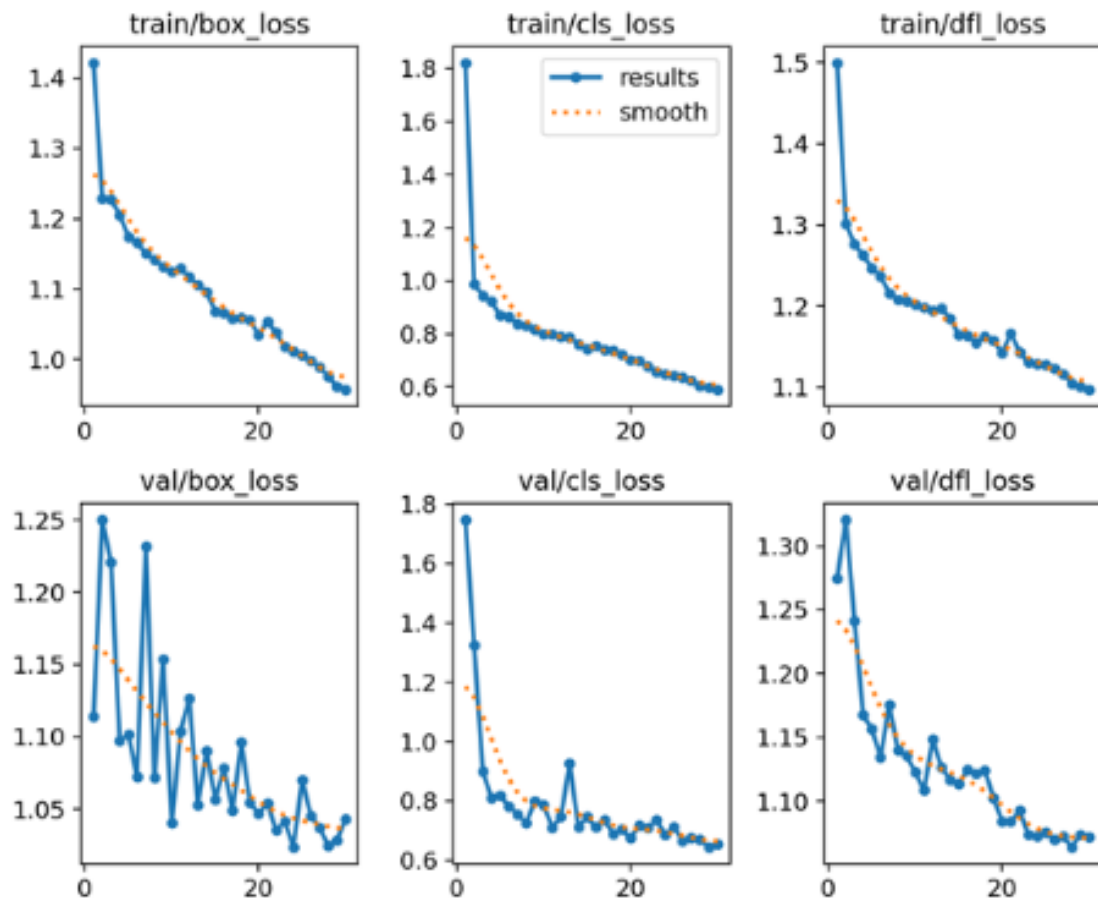


Figure 3: Blood cells detected image

9.2 Train and Validation Loss Graph



10 Streamlit Blood cellcount Application Overview Code Implementation

10.1 Step 1: Importing Required Libraries

The necessary libraries for running the Streamlit application, image processing, and model inference are imported as follows:

```
import streamlit as st
from PIL import Image
from ultralytics import YOLO
from collections import Counter
import os
```

10.2 Step 2: Load YOLO Model

We load the pre-trained YOLOv8 model, which is capable of detecting blood cells in images.

```
# Load the YOLO model
model = YOLO["best.pt"]
```

10.3 Step 3: Set App Title and Description

The app will have a clear title and description for user understanding.

```
st.set_page_config(layout="wide") # Enable wide mode for more horizontal space
st.title("🩸 Blood Cell Detection with YOLOv8")
st.write("Upload a microscopy image. The model will detect and count blood cells.")
```

10.4 Step 4: Upload Image

The user can upload images in either JPEG, PNG, or JPG format. The code for the file uploader is as follows:

```
# File uploader
uploaded_file = st.file_uploader("Upload Image", type=["jpg", "jpeg", "png"])

if uploaded_file:
    image = Image.open(uploaded_file).convert("RGB")
```

10.5 Step 5: Perform Inference

After the image is uploaded, we perform inference using the YOLOv8 model. The detection results will be visualized on the image.

```
# Run YOLOv8 inference
try:
    results = model.predict(image)
    results = results[0]
    result_img = results.plot()

    # Extract class IDs and names
    class_ids = re.findall(r'class_id: (\d+)', results.names, astype=int)
    class_names = results.names
    counts = Counter(class_ids)

    # Prepare output count dictionary
    blood_cell_count = {class_names[cls_id]: count for cls_id, count in counts.items()}
```

10.6 Step 6: Display Detection Image

Next, we display the image with the detected blood cells.

```
with col2:
    st.subheader("🖼️ Output (Colored Detections)")
    st.image(result_img, use_container_width=True)
```

10.7 Step 7: Display Summary Results

Finally, we display the summary of the detection, including the count of blood cells.

```
if blood_cell_count:
    st.markdown("### 📊 Blood Cell Count:")
    for cell_type, count in blood_cell_count.items():
        st.write(f"- **{cell_type}**: {count}")
```

11 Results and Discussions

11.1 Input Microscopic Blood Image

When the input microscopic blood image is provided to the model, it will display as shown in **Figure 4**.

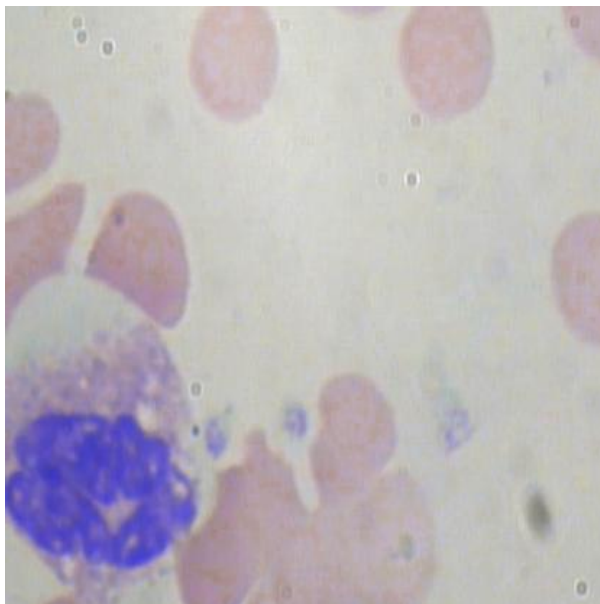


Figure 4: Uploaded Image

11.2 Detection and Highlighting Blood Cells

After processing the uploaded microscopy image, the model highlights all detected cells directly on the image. Different cell types (such as red blood cells and white blood cells), are marked.

This annotated detection image, shown in **Figure 5**, provides a clear visual understanding of where and what types of cells have been identified.

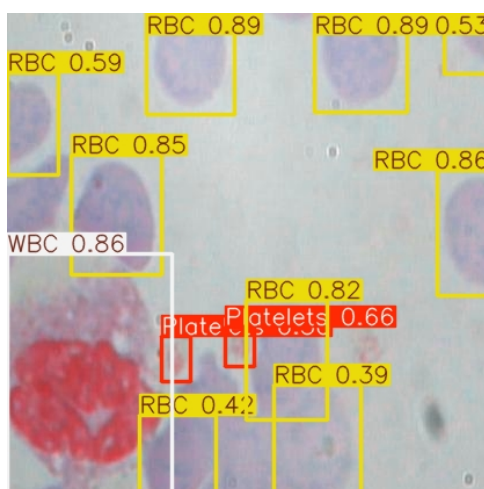


Figure 5: Summary of Detection

11.3 Summary of Detection

After analyzing the uploaded image, the model counts how many cells of each blood cell type were detected. The summary section displays the names of the blood cell types found along with the number of cells detected for each type.

11.4 Performance Metrics

Figure 6 below shows a table that summarizes the performance of the model based on three important metrics:

Insights from the Performance Metrics:

- The model detects red blood cells and leukocytes with high accuracy, achieving precision and recall values above 96
- Trophozoites and rings are also detected well, but slightly lower precision and recall values indicate some room for improvement.
- Detection of schizonts and difficult cells is relatively harder, with lower precision and recall values compared to other cell types.

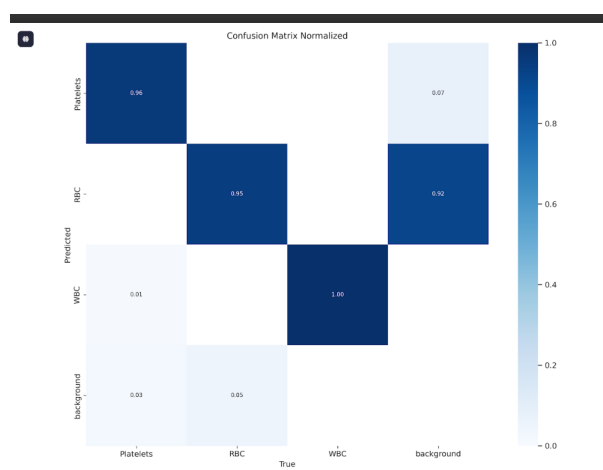


Figure 6: Performance Metrics

11.5 Confusion Matrix

The confusion matrix shown in **Figure 7** below tells how well the YOLOv8 model correctly classified each type of cell. Each row represents the actual class, and each column represents the predicted class.

Insights from the Confusion Matrix:

- Red blood cells and leukocytes are very accurately detected, with the diagonal cells (actual vs predicted) showing minimal errors.
- Trophozoites, rings, schizonts, and difficult classes show some confusion, especially among each other.
- Most errors occur between trophozoites, rings, and difficult cells, which are visually quite similar.

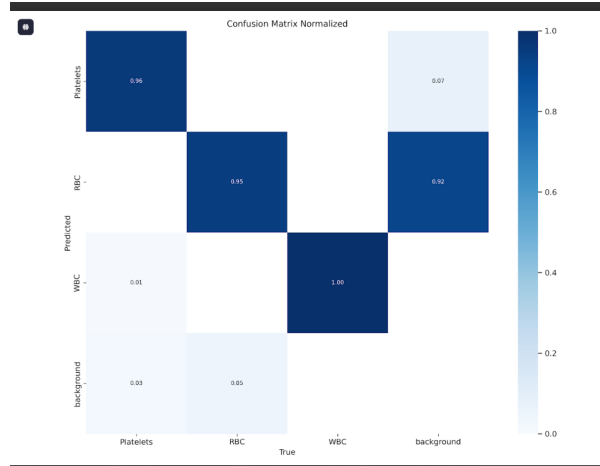


Figure 7: YOLOv8 Confusion Matrix

12 Conclusion

12.1 Summary of App's Contributions

The development of an application for predicting RBC, WBC, and platelets count using microscopic blood images with the YOLO model represents a significant advancement in the field of medical diagnostics. This app addresses the limitations of manual blood cell counting by leveraging state-of-the-art object detection algorithms to provide quick, accurate, and automated analysis. By utilizing the YOLO model, the app enhances the efficiency and accuracy of blood cell counting, thereby reducing human error and improving the overall diagnostic process.

12.2 Advancements over Traditional Methods

The app builds upon the foundational principles of traditional manual counting methods and automated systems, introducing several key improvements:

- **Enhanced Accuracy and Precision**
- **Increased Speed and Efficiency**
- **User-Friendly Interface**

12.3 Future Potential in Medical Diagnostics and AI Applications

The advancements introduced by this app open up exciting possibilities for future research and applications in medical diagnostics and artificial intelligence: **Medical Diagnostics:** The app's ability to provide rapid and accurate blood cell counts can be expanded to other diagnostic applications, potentially integrating with broader health monitoring systems. **AI Applications:** The success of using the YOLO model in this context paves the way for further AI-driven innovations in medical imaging and diagnostics, encouraging the development of more sophisticated tools for various medical fields. **Research and Development:** Continuous improvements and updates to the app can incorporate the latest

advancements in AI and machine learning, ensuring it remains at the forefront of medical technology and contributes to ongoing research efforts.

- **Medical Diagnostics:** The app's ability to provide rapid and accurate blood cell counts can be expanded to other diagnostic applications, potentially integrating with broader health monitoring systems.
- **AI Applications:** The success of using the YOLO model in this context paves the way for further AI-driven innovations in medical imaging and diagnostics, encouraging the development of more sophisticated tools for various medical fields.
- **AI Applications:** Continuous improvements and updates to the app can incorporate the latest advancements in AI and machine learning, ensuring it remains at the forefront of medical technology and contributes to ongoing research efforts.

13 Future Enhancements

13.1 Areas for Future Research and Development

The development of the application for predicting RBC, WBC, and platelets count using microscopic blood images with the YOLO model is a significant step forward. However, there are several areas where further research and development can enhance its capabilities: **Advanced Image Preprocessing Techniques:** Investigating more sophisticated image preprocessing methods could improve the accuracy and robustness of the model. Techniques such as advanced noise reduction, image enhancement, and normalization could help the YOLO model perform better under varied imaging conditions. **Integration with Other AI Models:** Combining the YOLO model with other machine learning or deep learning models could improve overall performance. For example, integrating with models specifically designed for segmentation, such as U-Net, could enhance the identification and counting of overlapping cells. **User Interface and Experience Enhancements:** Improving the app's user interface and experience can make it more accessible and user-friendly for medical professionals. This includes better visualization tools, more intuitive controls, and enhanced reporting features that provide comprehensive insights into the blood cell counts.

- **Advanced Image Preprocessing Techniques:** : Investigating more sophisticated image preprocessing methods could improve the accuracy and robustness of the model. Techniques such as advanced noise reduction, image enhancement, and normalization could help the YOLO model perform better under varied imaging conditions.
- **Integration with Other AI Models:** Combining the YOLO model with other machine learning or deep learning models could improve overall performance. For example, integrating with models specifically designed for segmentation, such as U-Net, could enhance the identification and counting of overlapping cells.
- **User Interface and Experience Enhancements:** Improving the app's user interface and experience can make it more accessible and user-friendly for medical professionals. This includes better visualization tools, more intuitive controls, and enhanced reporting features that provide comprehensive insights into the blood cell counts.

13.2 Potential Applications and Extensions

The application for predicting RBC, WBC, and platelets count using microscopic blood images with the YOLO model presents numerous potential applications and extensions across various fields:

Automated Medical Diagnostics: This app can be integrated into clinical workflows to automate blood cell counting, reducing the workload for laboratory technicians and increasing efficiency in medical diagnostics. It can provide rapid, accurate counts that aid in diagnosing conditions such as anemia, infections, and hematologic diseases. **Integration with Electronic Health Records (EHR):** The app can be linked with EHR systems to automatically update patient records with blood cell counts, streamlining the process of maintaining accurate and up-to-date medical records. **Enhanced Research**

Capabilities: Researchers can use the app to analyze large datasets of blood images, facilitating studies on blood cell morphology, disease progression, and the effectiveness of treatments. This can lead to new insights and advancements in hematology research.

- **Automated Medical Diagnostics:** This app can be integrated into clinical workflows to automate blood cell counting, reducing the workload for laboratory technicians and increasing efficiency in medical diagnostics. It can provide rapid, accurate counts that aid in diagnosing conditions such as anemia, infections, and hematologic diseases.
- **Integration with Electronic Health Records (EHR):** The app can be linked with EHR systems to automatically update patient records with blood cell counts, streamlining the process of maintaining accurate and up-to-date medical records.
- **Enhanced Research Capabilities:** Researchers can use the app to analyze large datasets of blood images, facilitating studies on blood cell morphology, disease progression, and the effectiveness of treatments. This can lead to new insights and advancements in hematology research.

References

- [1] Cruz D., Jennifer C., Valiente L.C., et al.: *Determination of blood components (WBCs, RBCs, and platelets) count in microscopic images using image processing and analysis*. Proc. Int. Conf. on Humanoid, Nanotechnology, Information Technology, Communication and Control, Environment and Management (HNICEM), December 2017
- [2] Acharjee S., Chakrabartty S., Alam M.I., et al.: A semiautomated approach using GUI for the detection of red blood cells. Proc. Int. Conf. on Electrical, Electronics, and Optimization Techniques, 2016, pp. 525–529
- [3] Islam M.T., Aowal M.A., Minhaz A.T., et al.: Abnormality detection and localization in chest X-rays using deep convolutional neural networks, arXiv preprint arXiv:1705.09850v3, 2017
- [4] Acharya V., Kumar P.: ‘Identification and red blood cell automated counting from blood smear images using computer-aided system’, *Med. Biol. Eng. Comput.*, 2018, 56, (3), pp. 483–489 (doi: 10.1007/s11517-017-1708-9) [PubMed] [Google Scholar]
- [5] Zhao J., Zhang M., Zhou Z., et al.: ‘Automatic detection and classification of leukocytes using convolutional neural networks’, *Med. Biol. Eng. Comput.*, 2017, 55, (8), pp. 1287–1301 (doi: 10.1007/s11517-016-1590-x)